Proposed AST-Route Four-Layer Multi-Objective Query Routing Framework

LAYER 1A: Input SQL & Static AST Parsing (SQLGlot)

- Input Workload: 75 Factorial SQL Queries (25 Canonical Seeds x V1-V3)
- Zero-Execution Static Parsing via SQLGlot (< 0.8 ms overhead)
- Extracts Operator Topology: Depth D(q), Nodes N(q), Joins J(q), Subqueries/CTEs S(q), Windows W(q), UNNEST U(q), CASE C(q)

LAYER 1B: 9-Dimensional Feature Vector & CSCI

- Validated Composite Index: CSCI(q) = 1.5D + 2.5J + 3.0S + 3.5W + 4.0U + 1.5C
- Construct Validity: Spearman rho = 0.4935, Weight Stability = 0.8189
- Zone-Map Predicate Selectivity Factor: sigma(q) in (0, 1] across V1-V3
- 9D Routing Vector: x_q = [D, N, J, S, W, U, C, CSCI, sigma(q)]^T in R^9

LAYERS 2 & 3: Cross-Paradigm Analytical Cost Models & Multi-Output Cost-Regression Engine (R_GBR-9D)

- Analytical Models: PCIe Fraction Phi_PCIe(q), Effective Throughput Theta_eff (GB/s), Cloud Slot Multiplier M_slot, FinOps Savings Delta C_1M ($)
- Multi-Output GBR Predictor f_Theta: R^9 -> R^M simultaneously estimates Log-Latency ln(1+T_e), Energy E_e (Joules), and Memory (RAM/VRAM GB)
- Out-of-Fold Predictive Accuracy (25-Seed LOSO-CV): GPU Latency R^2 = 0.875 | Polars Latency R^2 = 0.822 | Energy R^2 = 0.815 (Inference: 497.2 us)

Candidate 1: DuckDB (C++)

- Paradigm: In-Process CPU DBMS
- Morsel-Driven Vectorized Engine
- Segment-Tree Window Framing
- Mean Latency: 183.74 ms
- Energy: 6.40 J (243.0 MB/J)
- Routed: 80% L1, 20% L2

Candidate 2: Polars / DataFusion

- Paradigm: Rust Apache Arrow
- Zero-Copy SIMD & ListArray
- Polars: 82.47 ms (3.17 J)
- DataFusion: 96.30 ms (3.39 J)
- 102.1x Faster than Row-Store
- Routed: 60% L2 (Narrow/Med)

Candidate 3: CUDA GPU (SIMT)

- Paradigm: RTX 2050 (4GB VRAM)
- Cold: 56.96 ms | Warm: 22.26 ms
- L1 PCIe Bottleneck: Phi = 55.5%
- Energy: 0.65 J Cold / 0.25 J Warm
- Green Perf: 1942.8 MB/J Warm
- Routed: 60% L3, 100% L4, 80% L5

Candidate 4: Google BigQuery

- Paradigm: Serverless Cloud OLAP
- Capacitor Storage + Slot Elasticity
- Clustered Mean: 745.39 ms
- Slot Multiplier: M_slot = 6.37x
- V1 I/O Pruning: 78.4% ($1.47/1M)
- Elastic Out-of-Core Fallback

LAYER 4: Policy-Driven Optimal Engine Recommendation & Leak-Free 25-Seed LOSO-CV Performance

- Optimal Dispatch Rule: e_hat(q) = argmin_e [E(T_e | x_q)] -> Avoids L1/L2 PCIe Bottlenecks (0% L1 GPU) while Offloading L3-L5 to CUDA GPU
- Outperforms All Static & Heuristic Baselines: 54.92 ms Mean Latency (95% CI: [42.78, 68.50] ms), 34.98% Regret, 65.33% Near-Optimal Hit, 1.381 J Energy